

KRUPA PATEL, EIT

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PROFESSIONAL SUMMARY

Electrical and Electronics Engineer with 4+ years of experience in **electrical systems design, embedded systems development, testing, automation, 2D and 3D modeling and system integration**. Strong background in **hardware-software integration, PLC, PCB & HMI programming, firmware development, and electronic product testing**.

Experienced in working with cross-functional teams in manufacturing industry and aerospace environments, delivering reliable and efficient engineering solutions within deadlines.

TECHNICAL SKILLS

- Programming Language: C, C++, Python, SQL, FPGA Basics
- Testing & Measurement Tools: Oscilloscope, Spectrum Analyzer, DSO, Signal Generators, Multimeter etc.
- Automation & Industrial Systems: PLC programming, HMI development, SCADA basics, etc.
- Design & Engineering Tools: AutoCAD, Autodesk Inventor, Altium Designer, MATLAB, NI Multisim, Cadence, Xilinx ISE/Vivado, Keil, LabVIEW
- Additional Tools: Tableau, Power BI, Linux, Canva, MS Office Suite

PROFESSIONAL EXPERIENCE

Electrical Engineer

HeatMasterSS | March 2023 – Present

- Program and validate PLC logic, HMI interfaces, and PCB control systems using VS Code, C/C++.
- Conduct Streamlined mechanical component creation by building reusable Inventor iParts and iAssemblies.
- Translated engineering concepts and rough hand sketches into precise 3D digital prototypes and 2D manufacturing drawings. Design and draft comprehensive 2D electrical schematics in AutoCAD, mapping out power distribution, control loops, and sensor wiring for biomass furnace systems.
- Collaborate with manufacturing/production teams to resolve quality issues and support continuous process improvement. Maintain detailed test documentation, defect reports, validation records, and engineering change documentation.
- Analyze test data using Python, Excel, Power BI, and LabVIEW to identify trends and improve product quality. Develop and maintain Standard Operating Procedures (SOPs), test procedures, and technical documentation.
- Participate in continuous improvement initiatives using Lean and Six Sigma methodologies.

Electrical & Electronics Engineer

Space Applications Centre (ISRO) | Nov 2021 – Dec 2022

- Worked on SATCOM terminal systems and baseband signal processing applications
- Developed firmware for RF front-end configuration and NavIC data processing systems
- Implemented communication protocols including COSPAS-SARSAT standard formatting
- Performed firmware development and testing using MSP430 microcontroller, Code Composer Studio (CCS), and IAR Workbench. Designed MATLAB simulations for signal processing and communication system validation
- Integrated communication interfaces using UART, SPI, and I2C protocols

Engineering Technologist / Jr. Test Engineer

Prima Automation | Oct 2019 – Nov 2021

- Conducted functional testing and troubleshooting of electronic and electrical assemblies
- Interpreted engineering drawings, schematics, and technical documentation
- Performed assembly and testing of electronic components including resistors, capacitors, ICs, and transistors
- Developed AutoCAD-based electrical schematics, layouts, and single-line diagrams
- Supported firmware maintenance and debugging for legacy embedded systems
- Executed test plans, recorded results, and ensured compliance with quality standards
- Collaborated with engineering teams to meet project deadlines and specifications

EDUCATION

Bachelor of Technology (B.Tech) – Electronics and Communication Engineering

Relevant Coursework:

Embedded Systems, VLSI Design, Digital Signal Processing, Digital Communication, Power Electronics, Analog Electronics, Radar Engineering, Antenna & Propagation, Microcontrollers & Interfacing, Electronics Instrumentation, Image Processing

CERTIFICATIONS

- Engineering Intern of Engineers Geoscientists Manitoba
- AutoCAD Micro-Credential Program
- Electronics Circuits in Maritime Communication System
- LabVIEW Programming
- Python Programming Diploma
- PLC Developer Certification
- IoT & Embedded System
- Linux Fundamentals

PROJECTS

- Developed UHF Beacon Transmitter for GAGANYAAN mission (ISRO-related indigenous project)
- Contributed to Emergency Locator Transmitter (ELT) and Distress Alert Transmitter (DAT) systems development
- Programmed proprietary in-house control boards and software, introducing real-time diagnostic reporting and automatic over-the-air (OTA) updates for consumers.
- Collaborated on the development of the myHeatMaster app, syncing physical EZ Link Control hardware with cloud-based user alerts and remote temperature monitoring.
- Designed and optimized high-efficiency combustion/gasification systems.
- Built Bluetooth-based Home Automation System using embedded controllers
- Developed Solar Tracking System using Arduino and sensor integration
- Created Motion Detection Alert System using Arduino and Python SMS integration
- Built FPGA-based digital filter design and verification system
- Developed Raspberry Pi-based Smart Security System with sensor integration